



University of Stuttgart
Germany



Herman Hollerith Zentrum

Distributed Systems

Overview and Logistics

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Summer

Topics

- Architecture models
- Sockets and multiprocessing
- Dynamic discovery
- Multicast
- Logical time
- Leader election
- Reliable ordered multicast

What to expect?

- Discuss system design decisions
- How-to of selected concepts
- Examples
- Pointers to tutorials

Tentative schedule

Date	Topic
11.05.26	Overview and logistics + core design challenges + client-server arch. model
12.05.26	Client-server architecture model (cont.) + dynamic discovery
01.06.26	Dynamic discovery (cont.) + logical time + leader election
02.06.26	Leader election (cont.) + reliable ordered multicast

What do you have to do?

STUDENT PROJECT

Objective

Design and implementation of a
distributed system

Your project idea

Develop a project idea with distribution properties

LOGISTICS

Groups

- Groups of three students
 - Self-organising and self-managing
 - Collaborate with each other (e.g., GitHub, Slack, Trello)
 - Distribute work equally
 - Everyone contributes
- Each group gets a number
 - Use that number as ID for your group



Project requirements

- Architectural model
- Dynamic discovery of hosts
- Voting
- Fault tolerance

Deliverables

- Preliminary Project Form
- Updated Project Form (Optional)
- Project Report Form
- Project Demo

Project report form

- Sections
 - Project Information
 - Architectural Model
 - Dynamic Discovery of Hosts
 - Voting (Leader Election)
 - Fault Tolerance
- Instructions
 - Complete **all sections** of the form.
 - Adhere to the **character limits** in text fields.
 - Ensure text is **clear** and **readable** in all boxes.
 - Export your final submission as a **non-editable PDF**.
- Template
 - PDF form available on ILIAS/RELAX
- Submission format
 - **Non-editable PDF**

Project demo

- 15 minutes per group
- Be prepared
- Demonstrate the working of your system
- Q&A
- Schedule will be announced in due course

How to submit?

- RELAX

About projects

FAQ

Do we have to use a specific programming language?

No. You may use any programming language(s) of your choice.

Can we use an existing middleware (e.g., Zookeeper)?

No. If a middleware already implements at least one of the project requirements, it is not allowed to be used for your project.

Can we demonstrate our distributed system on one computer?

No. It is not possible to demonstrate distribution on a single computer. Your system must run on **at least two physical machines** during the demo.

Do we need to develop a graphical user interface?

No. A graphical user interface is not required.

Can/Should we put code in the final report?

No. Do **not** include code in the report. Instead, provide a link to your code repository (e.g., GitHub, Bitbucket, etc.). If your repository is private, add **Ilche** as a collaborator.

My GitHub/Bitbucket username: ilche

My GitLab username: ilche.georgievski

My TIK GitHub username: contact me via email

Do you provide programming/debugging support?

No. We do not provide programming/debugging assistance.

Can you help resolve group conflicts?

No. We do **not** get involved in group dynamics. Groups are expected to be **self-organising** and **self-managing**.

Can we use Generative AI for creating project deliverables?

You can use whatever Generative AI tool you want for the deliverables, but **you** are solely responsible for the content. Hallucinations, incorrect statements, and faulty code are **your** mistakes and can lead to failing the project.